

Bhojpuri Whisper AI: Training & Evaluation Report

A Beginner-Friendly Analysis of Speech Recognition Fine-Tuning & WER Progression

1. Executive Summary & Key Highlights

This report analyzes the fine-tuning of OpenAI's **Whisper-Small** on rural Bhojpuri voice data (ai4bharat/Rural_Women_Bhojpuri). Training was conducted over 7 sessions reaching Step 15,500 (~1.48 Epochs).

- **Base Model:** openai/whisper-small (240 Million Parameters)
- **Target Training:** 3.0 Full Epochs (31,443 Total Steps)
- **Current Progress:** Step 15,500 (~1.48 Epochs completed, ~49.3% of total goal)
- **Starting Accuracy:** 71.50% Word Error Rate (WER) at Step 100
- **■ All-Time Best Accuracy:** 40.58% Word Error Rate (WER) at Step 14,900
- **Total Evaluations:** 149 validation checkpoints evaluated
- **Best Model Location:** models/bhojpuri-whisper-small-full/checkpoint-14900

2. Beginner's Guide: Understanding the Terminology

Term	Simple Meaning	Everyday Analogy
WER (Word Error Rate)	Percentage of words the AI got wrong. Lower is better.	If AI hears 100 words and gets 40 wrong, WER is 40%.
Eval Loss	A math score for AI mistakes/uncertainty. Lower is better.	Like marks deducted on a test. Lower loss = fewer errors.
Training Step	One iteration where AI practices on a batch of audio files.	Solving 1 homework problem.
Epoch	One complete pass through the entire dataset from start to finish.	Reading a textbook from cover to cover one full time.
Checkpoint	A saved snapshot of the AI's memory at a specific step.	Saving game progress so you can resume anytime.

3. Training Milestones & WER Progression

Milestone	Step	Epoch	WER (%)	Eval Loss
Starting Point	100	0.01	71.50%	0.694
Early Learning	400	0.04	58.88%	0.435
Initial Plateau	1,100	0.10	51.21%	0.355
Full Epoch 1	10,800	1.03	42.34%	0.280
■ All-Time Record	14,900	1.42	40.58%	0.265
Current State	15,500	1.48	43.80%	0.264

4. Top 5 Best Performing Checkpoints

Rank	Checkpoint Step	WER (%)	Eval Loss	Saved on Disk?
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■ 1st Place	Step 14,900	40.58%	0.2655	■ Yes (checkpoint-14900)
■ 2nd Place	Step 14,500	41.06%	0.2604	Rotated out
■ 3rd Place	Step 15,200	41.30%	0.2626	Rotated out
4th Place	Step 13,900	41.57%	0.2643	Rotated out
5th Place	Step 10,800	42.34%	0.2803	■ Yes (checkpoint-10800)

5. Practical Insights & Recommended Next Steps

- **Linguistic Accuracy:** Dropping WER from 71.5% to 40.58% represents a ~43% relative error reduction, significantly improving Devanagari recognition of authentic rural Bhojpuri speech.
- **Model to Use Now:** **checkpoint-14900** is the optimal checkpoint for transcriptions and production tests.
- **Resume Training:** To finish the remaining ~15,943 steps (reaching 3 full epochs), update resume.bat to point to checkpoint-15500.